# Phase 2: RFM Customer Segmentation (Summary)

In this phase, I used clean data to build useful business intelligence by applying the RFM (Recency, Frequency, Monetary) model. Olist is a marketplace platform that connects independent sellers with buyers — similar to how Allegro operates in Poland. The main goal was to understand how customers behave across the platform and identify the most valuable segments to help Olist make better strategic decisions.

## 1. Methodology & Research Process

**Technical Challenges:** Early results suggested almost no returning customers. I found that this was because the customer_id was unique to each order session. To correctly identify repeat buyers, I switched to using customer_unique_id as a persistent identifier.

**Error Elimination:** I found a row inflation problem in the payments table. Orders paid in installments created up to 29 rows per order, which made the number of transactions look much higher than it really was. I used COUNT(DISTINCT order_id) to get the correct count.

**SQL Implementation:** The model scores customers on two dimensions — Recency (how many days since the last purchase, scored 1 to 5) and Monetary (total amount spent, scored 1 to 5). Frequency was left out on purpose because most Olist customers buy only once, which makes frequency a poor way to separate them into groups.

## 2. Segment Definitions

Five segments were created based on combinations of Recency and Monetary scores:

**Recent Champions (R=5, M=5):** Customers who bought very recently (within the last 90 days) and spent over 1,000 R$. This is the highest-value group.

**At Risk High-Value (R<=2, M=5):** Customers who spent over 1,000 R$ in total but have not bought anything in over 270 days. Their revenue is at risk of being lost.

**New / Active (R>=4):** Customers who bought within the last 180 days, regardless of how much they spent. This is the most recently active group.

**Lost Low-Value (R<=2, M<=2):** Customers who have not bought in over 270 days and spent less than 50 R$. Low activity and low value.

**Occasional:** All customers who do not fit any other segment. They typically have a mid-range Recency and mid-range spend — not recently active enough or high-value enough to qualify elsewhere.

## 3. Key Business Insights

**The "Leaky Bucket" Challenge:** 97% of customers made only one purchase. This means the platform depends heavily on acquiring new customers rather than keeping existing ones — a costly strategy that is difficult to sustain long term.

**Hidden Loyalty:** Despite the low overall retention, there are 2,801 repeat customers across all segments — mainly in New/Active (1,213) and Occasional (1,211). The overall repeat rate is just 3%, which shows that Olist runs on a volume-driven model rather than a loyalty-based one.

**Key findings by segment:**

Recent Champions: 225 customers with a 10.7% retention rate (24 repeat buyers) — well above the platform average.

At Risk High-Value: 473 customers who generated 760,104 R$ in total (~1,607 R$ per customer). They make up only 0.51% of all customers but account for 4.93% of total revenue.

New/Active vs Occasional Paradox: Both segments generate almost the same total revenue (~6.08M R$ each), but Occasional customers spend an average of 210 R$ compared to 160 R$ for New/Active. This suggests that customers who return after a longer break tend to spend more per order.

## 4. Strategic Recommendations

**Platform Loyalty Programme:** The 3% repeat rate shows that Olist has no strong mechanism to bring customers back. Introducing a platform-wide loyalty programme — similar to Allegro Smart — could encourage repeat purchases across all sellers and increase the overall retention rate.

**Re-engagement Tools for Sellers:** Olist could give sellers access to tools that allow them to send personalised offers to customers who have not returned in over 90 days. The 473 At Risk High-Value customers represent 760K R$ in potential revenue — targeted re-engagement at the platform level could recover a significant part of this.

**Champion Segment Analysis:** The behaviours of the 24 repeat buyers in the Champions segment should be studied in more detail. Understanding which product categories or delivery standards drive their loyalty could help Olist set better standards across all sellers.

## 5. Power BI Dashboard

The RFM results were shown on a Customer Segments page in Power BI with the following key elements:

- **5 KPI cards:** Total Customers (93K), Repeat Customers (3K), At Risk High-Value (473), Repeat Rate % (3%), Revenue At Risk (760K R$). The At Risk cards are highlighted in red to show urgency.
- **Segment breakdown table** with a Revenue per Customer column. The At Risk row is shown in red using conditional formatting.
- **Revenue by Customer Segment** bar chart and a Customer Base by Segment donut chart to show the overall structure of the customer base.

## 6. Future Optimization

The RFM model was built entirely in SQL to keep the analysis fast and easy to maintain. In the future, there is potential to expand the Power BI dashboard to show how customers move between segments over time.